

Canonical Routing

With canonical routing,

$$p_i(b) = G_i(b),$$

agent i receives samples according to

$$B(b) G_i(b).$$

In `position_only`, the centroid weight is

$$w(b) = 1 - G_i(b).$$

Therefore the target centroid is

$$\hat{x}_i = \frac{\sum_b B(b) G_i(b) (1 - G_i(b)) b}{\sum_b B(b) G_i(b) (1 - G_i(b))}.$$

The update rule is

$$x_i^+ = (1 - \beta)x_i + \beta\hat{x}_i.$$

Hence

$$\Delta x_i = x_i^+ - x_i = \beta \frac{\sum_b B(b) G_i(b) (1 - G_i(b)) (b - x_i)}{\sum_b B(b) G_i(b) (1 - G_i(b))}.$$

This matches the gradient direction up to:

1. left multiplication by the Σ -metric (since the exact gradient contains Σ^{-1}),
2. a positive scalar normalization

$$\sum_b B(b) G_i(b) (1 - G_i(b)),$$

3. step size β .

So for isotropic Σ , this is essentially a preconditioned/scaled gradient step, which is why the update is called *gradient-matched in direction*.